Group probability table - probability of failure over multiple instances

DC	% Chance of Failure	Exactly 1	Exactly 2	Exactly 3	Exactly 4	At LEAST 1	At LEAST 2
1	0%	0%	0%	0%	0%	0%	0%
2	5%	17%	1%	0%	0%	19%	1%
3	10%	29%	5%	0%	0%	34%	5%
4	15%	37%	10%	1%	0%	48%	11%
5	20%	41%	15%	3%	0%	59%	18%
6	25%	42%	21%	5%	0%	68%	26%
7	30%	41%	26%	8%	1%	76%	35%
8	35%	38%	31%	11%	2%	82%	44%
9	40%	35%	35%	15%	3%	87%	52%
10	45%	30%	37%	20%	4%	91%	61%
11	50%	25%	38%	25%	6%	94%	69%
12	55%	20%	37%	30%	9%	96%	76%
13	60%	15%	35%	35%	13%	97%	82%
14	65%	11%	31%	38%	18%	98%	87%
15	70%	8%	26%	41%	24%	99%	92%
16	75%	5%	21%	42%	32%	100%	95%
17	80%	3%	15%	41%	41%	100%	97%
18	85%	1%	10%	37%	52%	100%	99%
19	90%	0%	5%	29%	66%	100%	100%
20	95%	0%	1%	17%	81%	100%	100%

Understanding Probability in a Group Setting

The table above reveals a crucial insight about how probability works when multiple attempts are made. There are three key concepts at play:

1. Static (Individual) Probability - This is the simple "per roll" chance. For example: DC 11 gives each player a 50% chance to fail on their own check.
2. Binomial (Exact Count) Probability - This tells us the chance of getting a specific number of failures. With 4 players rolling:

Exactly 1 failure: 25% chance

Exactly 2 failures: 38% chance

Exactly 3 failures: 25% chance

Exactly 4 failures: 6% chance

Notice something important: the most likely outcome at DC 11 is 2 failures (38%), not 1 or 0. Even though each player individually has a coin-flip chance, the group as a whole will typically experience multiple setbacks.

3. Cumulative (At Least) Probability - This is the real eye-opener. It answers: "What are the odds that something bad happens this round?"

At DC 11: 94% chance at least one player fails each round

At DC 11: 69% chance at least two players fail each round

The Critical Insight

What feels "fair" for one player becomes a nearly certain failure for a group.

A DC that gives each player a "50/50 chance" (which seems balanced) actually creates a situation where:

15 out of 16 rounds will see at least one failure (94%)

2 out of 3 rounds will see two or more failures (69%)

This is why cascading penalties create death spirals: when failure is virtually guaranteed each round, and penalties make future rolls harder, the system quickly becomes mathematically unwinnable.

Practical Application
For sustainable tension:

DC 6-8 (25-35% individual failure) creates natural tension without spirals

DC 9-11 (40-50% individual failure) creates high-risk scenarios

DC 12+ (55%+ individual failure) almost guarantees multiple failures per round

The sweet spot for a challenging but fair hex-crawl is DC 6-9 (25-40% individual failure rate), which translates to:

68-87% chance of some setback each round

26-52% chance of multiple setbacks each round

Creates tension without mathematical certainty of collapse

All you need to do is find the target base DC that you want and then add +X to account for proficiency and attribute modifiers.

If the day still feels too easy, then you can increment by +1 DC midway through the adventuring day.

This explains why your playtest felt punishing: at DC 14 (effectively DC 11 for a typical party), you were operating in the “near-certain multiple failures per round” zone, making resource management feel impossible rather than strategic.